$$\frac{C}{2\pi} = r$$

$$r = \frac{132}{2 \times \frac{22}{7}}$$

$$= \frac{132}{2} \times \frac{7}{22}$$

$$= 21\, m$$

$$d = 2r$$

$$= 2 \times 21$$

$$= 42\ m$$

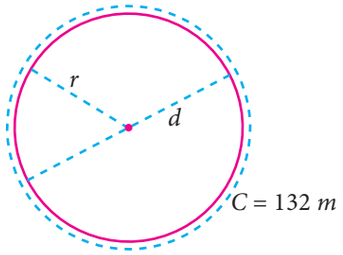


Fig. 2.6

Example 2.4 What is the distance travelled by the tip of the seconds hand of a clock in 1 minute, if the length of the hand is 56 *mm* (use $\pi = \frac{22}{7}$).


Fig. 2.7

Solution

Here the distance travelled by the tip of the seconds hand of a clock in 1 minute is the circumference of the circle and the length of the seconds hand is the radius of the circle. So, $r = 56\ mm$

Circumference of the circle, $C = 2\pi r\ units$

$$= 2 \times \frac{22}{7} \times 56$$

$$= 2 \times 22 \times 8$$

$$= 352\ mm$$

Therefore, distance travelled by the tip of the seconds hand of a clock in 1 minute is 352 *mm.*

Example 2.5 The radius of a tractor wheel is 77 *cm*. Calculate the distance covered by it in 35 rotations? (use $\pi = \frac{22}{7}$)


Fig. 2.8

Solution

The distance covered in one rotation

$$= \text{the circumference of the circle}$$

$$= 2\pi r\ units$$

$$= 2 \times \frac{22}{7} \times 77$$

$$= 2 \times 22 \times 11$$

$$= 484\ cm$$

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